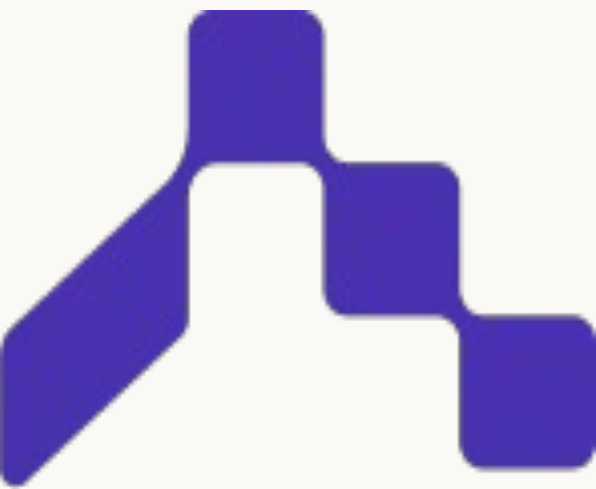




أكاديمية طويق
Tuwaiq Academy

Good To Go System

Description

The system is an online platform designed to reduce food waste by connecting restaurants ,bakeries and stores with customers

Users can discover restaurants, bakeries, and stores that have extra food available at discounted prices

They can choose an offer ,reserve it and pick it up at the specified time.

The system allows users to manage their reservations and orders ,save their favorite places and rate the food they receive

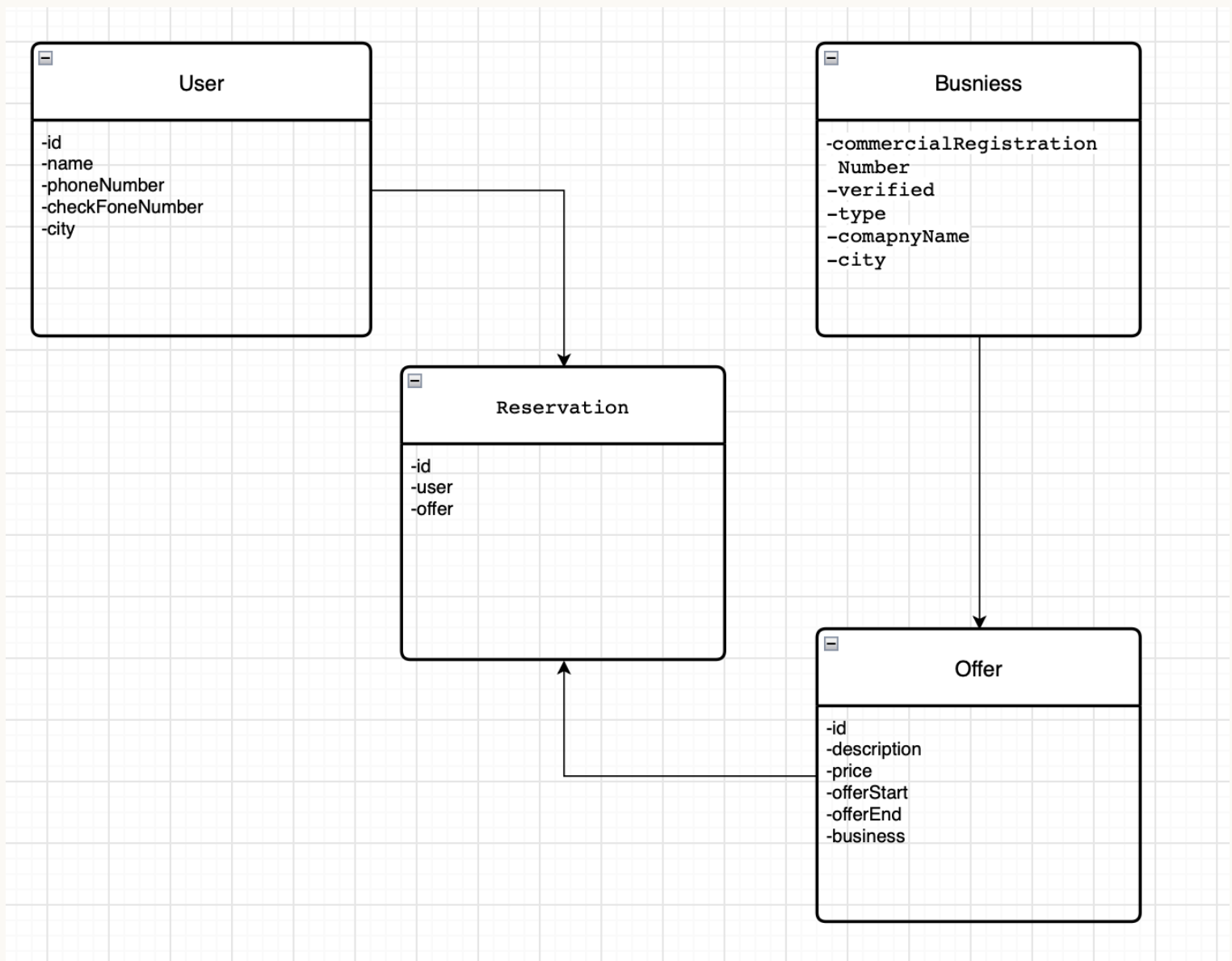
On the business side , restaurants and stores can add their available surplus food to the system ,set the price and quantity and manage their offers and reservations

Objectives

- Reduce food waste .
- Help restaurants and stores benefit from their surplus food.
- Provide customers with affordable food options.
- Support a more sustainable future

Class Relationship

- **Reservation is associated with a User** because every reservation must be made by a user.
- **Reservation is associated with an Offer** because every reservation must be for a specific offer.
- **Offer is associated with a Business** because each offer belongs to a specific business.



This creates the relationship: **User** → **Reservation** ← **Offer** ← **Business**.

Validation Checklist

User

Field	Validation Criteria	Potential Consequence of Failure
ID	Not Empty + exactly 10 digits	Invalid user identification
Name	Not Empty + letters only	Invalid user name
Phone Number	Not Empty + starts with 05 + exactly 10 digits	Invalid contact information
Phone Verification	Must be true	User cannot be verified
City	Not Empty + letters only + size 4–9 characters	Invalid location information

```
public class User { no usages

    @NotEmpty(message = "please Enter your id ") no usages
    @Pattern(regexp = "^\\d+$", message = "ID must be an integer")
    private String id;

    @NotEmpty(message = "Please Enter your name ") no usages
    @Pattern(regexp = ("^[a-zA-Z]$"))
    private String name ;

    @NotEmpty(message = "Please Enter your Phone Number ") no usages
    @Pattern(regexp = ("^05\\d{8}"),message = ("Please enter a valid phone number"))
    private String phoneNumber;

    @AssertTrue(message = "Sorry we need verify yor Phone Number") no usages
    private boolean checkPhoneNumber;

    @NotEmpty(message = "Please Enter your city") no usages
    @Pattern(regexp = ("^[a-zA-z]$"),message = ("City must contain letters only"))
    @Size(min = 4,max = 9)
    private String city;
```

Validation Checklist

Business

Field	Validation Criteria	Potential Consequence of Failure
Commercial Registration Number	Not Empty + exactly 10 digits	Invalid business identification
Verification	Must be true	Business cannot be verified
Type	Not Empty + must be restaurant, bakery, or store	Invalid business type
Company Name	Not Empty + size 2–16 characters	Invalid company name
City	Not Empty + letters only + size 4–9 characters	Invalid business location

```
public class Business { 1 usage

    @NotEmpty(message = "Please Enter the id") no usages
    @Pattern(regexp = ("^\\d{10}$"),message = ("please enter commercial Registration Number"))
    private String commercialRegistrationNumber;

    ⚡ @AssertTrue(message = "commercial Registration Number |Most be trou") no usages
    private boolean verified;

    @NotEmpty(message = "please Enter the type of your company") no usages
    @Pattern(regexp = ("^(restaurant|bakery|store)$")
    ,message = "Should select on of these restaurant or bakery or store")
    private String type;

    @NotEmpty(message = "Please Enter Company Name") no usages
    @Size(min = 2,max = 16)
    private String companyName;

    @NotEmpty(message = "Please Enter your city") no usages
    @Pattern(regexp = ("^[a-zA-Z ]+$"),message = ("City must contain letters only"))
    @Size(min = 4,max = 9)
    private String city;

}
```

Validation Checklist Offer

Field	Validation Criteria	Potential Consequence of Failure
ID	Not Empty + exactly 5 digits	Invalid offer identification
Description	Not Empty + size 10–100 characters	Invalid offer description
Price	Not Null + greater than 0	Invalid offer price
Offer Start	Must be today or in the future	Offer cannot be started
Offer End	Must be today or in the future	Offer cannot be ended
Business	Not Null	Offer cannot be associated with a business

```

public class Offer { 1 usage

    @NotEmpty(message = "Please Enter the offer id ") no usages
    @Pattern(regexp = ("^\\d{5}$"))
    private String id;

    @NotEmpty(message = "Please Enter the description ") no usages
    @Size(min=10,max = 100)
    private String description;

    @NotNull(message = "please Enter the offer price") no usages
    @Positive(message = "the price must be more thin 0 ")
    private Double price;

    @JsonFormat(pattern = ("yyyy-MM-dd")) no usages
    @FutureOrPresent(message = "start offer most be future or present ")
    private LocalDate offerStart;

    @JsonFormat(pattern = ("yyyy-MM-dd")) no usages
    @FutureOrPresent(message = "end offer most be future or present")
    private LocalDate offerEnd;

    @NotNull(message = " Enter Business ") no usages
    private Business business;

```

Validation Checklist

Reservation

Field	Validation Criteria	Potential Consequence of Failure
ID	Not Empty	Invalid reservation identification
User	Not Null	Reservation cannot be associated with a user
Offer	Not Null	Reservation cannot be associated with an offer

```
public class Reservation {  no usages

    @NotEmpty(message = "Enter the Reservation id ")  no usages
    private String id ;

    @NotNull(message = ("Enter the user to tick Reservation"))  no usages
    private User user;

    @NotNull(message = ("Most enter the offer "))  no usages
    private Offer offer;
```

Future Development

Market Inspiration

Existing applications in the Saudi market, such as **KNZ** and **Barakah** offer a similar concept by allowing users to benefit from surplus food at discounted prices. The idea of **Good To Go** was inspired by these applications with the goal of developing a more comprehensive user experience.

Proposed Features

- **Recommendations** — Suggest offers based on user preferences.
- **SmartNotifications & Promotions** — Notify users about new offers and promotions.
- **Ratings & Reviews** — Allow users to rate their experience.
- **Restaurant Reservations** — Enable users to reserve restaurants through the system
- **Location-Based Offers** — Show offers available near the user.

Summary

Good To Go is a simple system for managing **users, businesses, offers, and reservations**. It ensures accurate data through **validation** and can be enhanced with more features in the future.

Thank You